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WING STRUCTURE LAB

ERIC TRAN - GABRIEL KEITH CHOI

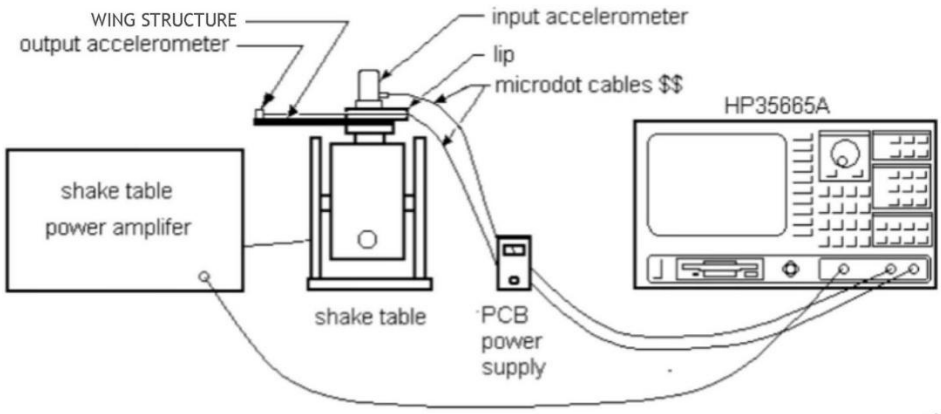
MOHAMMAD HASAN



LAB OBJECTIVES

- CONDUCT A SINE SWEEP AND RANDOM NOISE TEST ON A WING STRUCTURE MODEL WITH MINIATURE ENGINES (A COMPLEX-SHAPE BEAM WITH MULTIPLE MASSES).
- COMPARE THE PROCEDURE AND RESULTS WITH A CANTILEVER BEAM EXPERIMENT.
- EVALUATE THE ADVANTAGES AND DISADVANTAGES OF USING THE SINE SWEEP COMPARED TO THE RANDOM NOISE TEST.
- PRACTICE SETTING UP AND USING THE SPECTRUM ANALYZER TO CREATE FREQUENCY RESPONSE GRAPHS FOR DATA.
- EVALUATE THE EXPERIMENTAL NATURAL FREQUENCIES WITH THE THEORETICAL VALUES CALCULATED IN THE PRE-LAB.

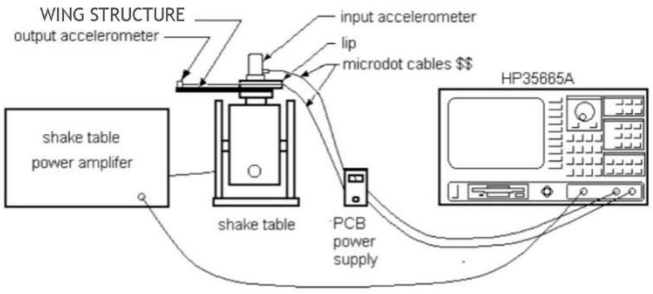
EXPERIMENTAL HARDWARE



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SINE SWEEP TEST (EXPERIMENTAL SETUP)

- INSTEAD OF A CANTILEVER BEAM, THE WING STRUCTURE MODEL IS ATTACHED TO THE PISTON OF THE SHAKE TABLE.
- ONE ACCELEROMETER IS PLACED AT THE CENTER OF THE WING AND THE OTHER IS ATTACHED TO THE END OF THE WING TO MEASURE VIBRATIONS.
- THE SPECTRUM ANALYZER GENERATES THE SINE WAVE INPUT AS WELL AS CAPTURE AND GRAPHS THE SYSTEM'S RESPONSE. IT WILL USE THE SWEPT SINE SETTING.
- MICRODOT AND BNC CABLES CONNECT THE ACCELEROMETERS TO THE SPECTRUM ANALYZER.



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SINE SWEEP TEST (FRF GRAPHS)



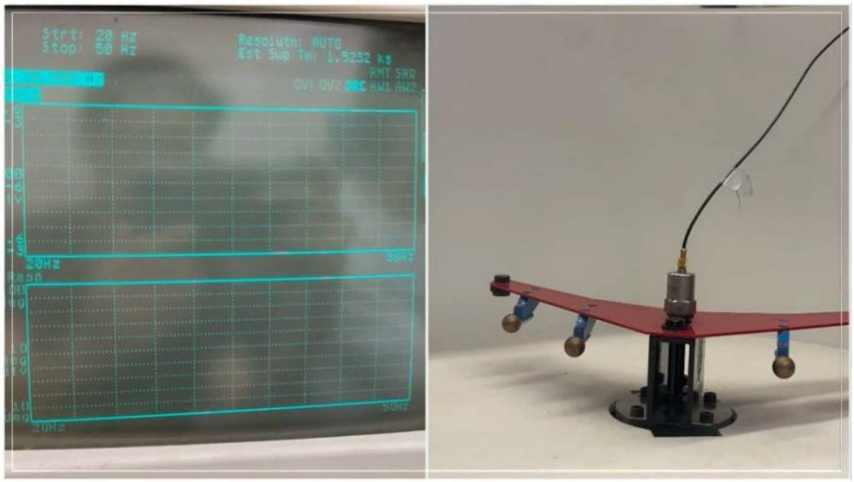


TEST 1 (10-310 Hz) RESULTS:
Wn1[Hz]: 31.25 Hz
Wn2[Hz]: 215.5 Hz

TEST 2 (20-50 Hz) RESULTS:
Wn1[Hz]: 31.475 Hz

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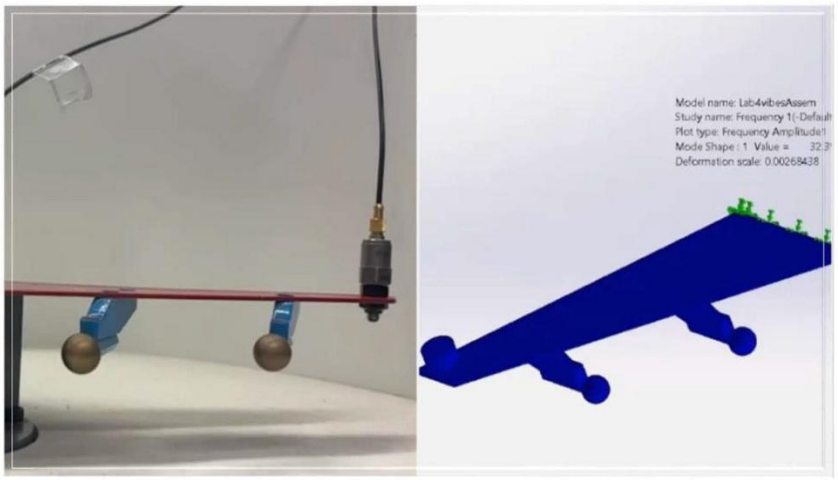
SINE SWEEP TESTING (FUNDAMENTAL NATURAL FREQUENCY)



FREQUENCY RESPONSE THROUGH THE BEAM'S NATURAL FREQUENCY

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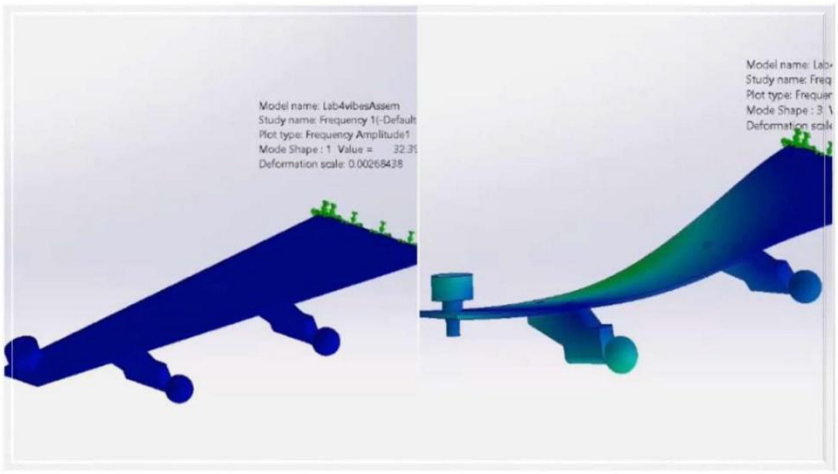
RESPONSE TO FORCED VIBRATION AT RESONANCE



FIRST RESONANT FREQUENCY VIBRATION IN SLOW MOTION

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MODE SHAPES OF WING STRUCTURE

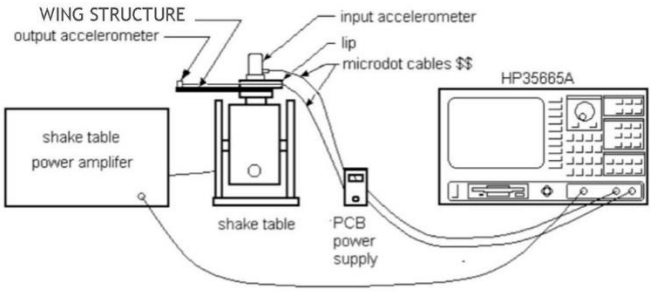


BENDING MODES 1 (LEFT) AND 2 (RIGHT) IN SOLIDWORKS

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RANDOM NOISE TEST (EXPERIMENTAL SETUP)

- THE SAME HARDWARE IS USED FOR THE RANDOM NOISE TEST (SEE SINE SWEEP TEST: EXPERIMENTAL SETUP)
- THE SPECTRUM ANALYZER'S SETTINGS ARE CHANGED TO PERFORM FFT ANALYSIS (RANDOM NOISE)



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RANDOM NOISE TEST (FRF GRAPHS)



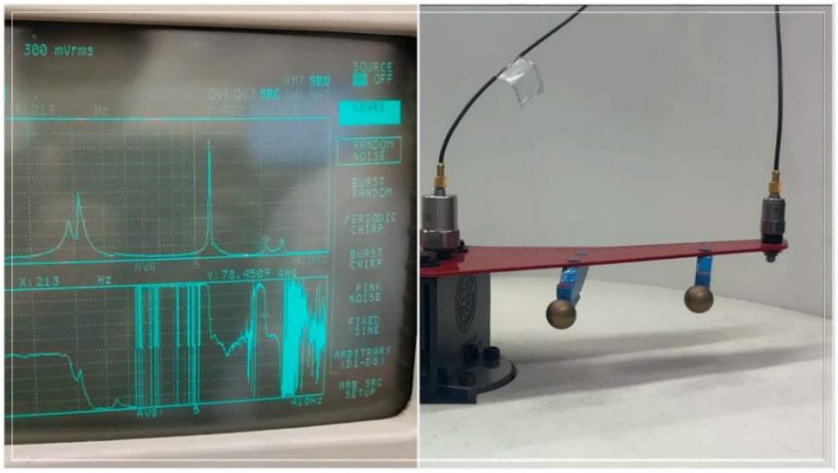


TEST 1 (10-310 Hz) RESULTS:
Wn1[Hz]: 31.5 Hz
Wn2[Hz]: 213 Hz

TEST 2 (20-50 Hz) RESULTS:
Wn1[Hz]: 31.5 Hz

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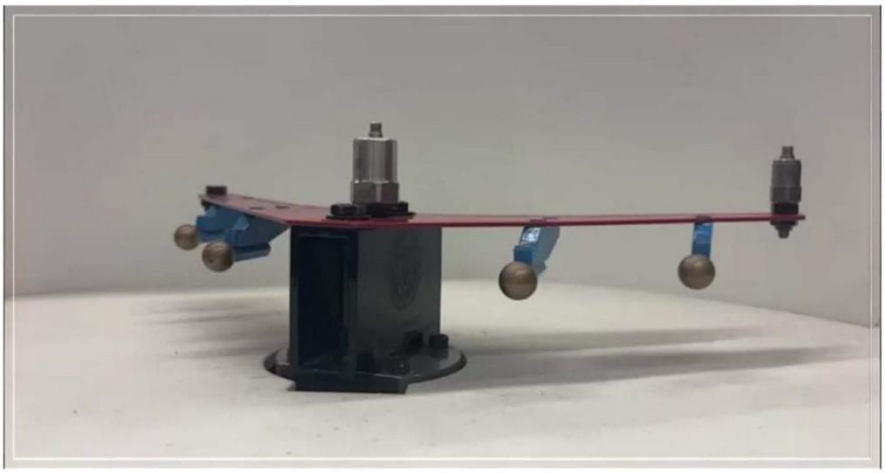
RANDOM NOISE TEST (FUNDAMENTAL NATURAL FREQUENCY)



FREQUENCY RESPONSE VIA THE RANDOM NOISE TEST

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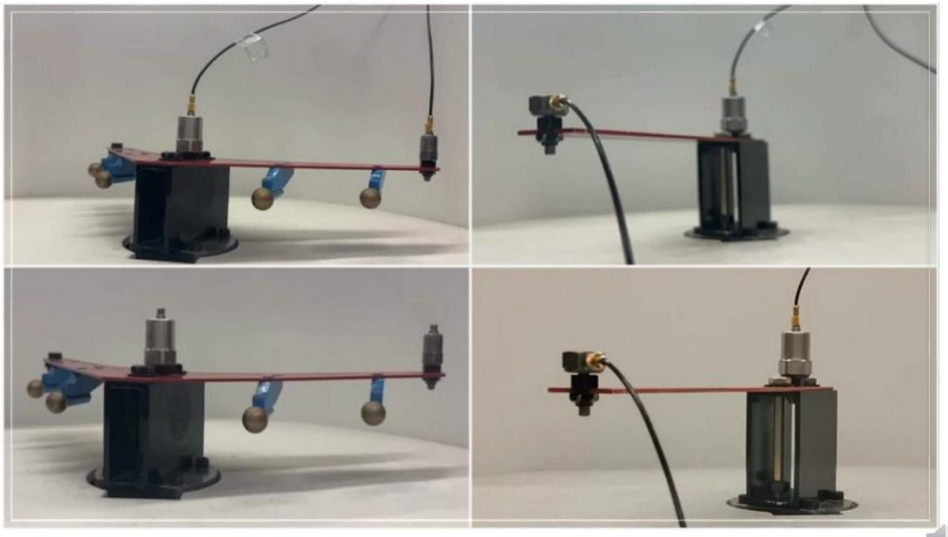
RANDOM NOISE RESPONSE OF WING STRUCTURE



RANDOM NOISE RESPONSE IN SLOW MOTION

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SLO-MO RESPONSE COMPARISON



WING AND BEAM FIRST RESONANT FREQUENCY VIBRATION (TOP ROW)

WING AND BEAM RANDOM NOISE EXCITATION (BOTTOM ROW)

ERROR ANALYSIS

TEST	Wn1(Hz)	ERROR(%)	Wn2(Hz)	ERROR(%)
CALCULATED VALUES	32.312	N/A	247.31	N/A
SINE SWEEP 1	31.25	3.3%	215.5	13%
SINE SWEEP 2	31.475	2.6%	N/A	N/A
RANDOM NOISE 1	31.5	2.5%	213	14%
RANDOM NOISE 2	31.5	2.5%	N/A	N/A

CONCLUSION AND DISCUSSION

- FOUND WN1 ERRORS UNDER 3.3%, BUT LARGER ERRORS (UP TO 14%) FOR WN2 DUE TO TEST LIMITATIONS. A SIMILAR ERROR AROSE IN THE CANTILEVER BEAM EXPERIMENT.
- THE COMPLEX, ASYMMETRICAL SHAPE OF THE WING RESULTS IN NATURAL FREQUENCIES FOR BENDING AND TWISTING MOTIONS WHEREAS THE CANTILEVER BEAM HAS ONLY BENDING BECAUSE OF ITS UNIFORM CROSS-SECTION.
- THE FIRST TWISTING MODE FOR THE BEAM OCCURS AT ROUGHLY 120 HZ (MEASURED BY SOLIDWORKS TO BE 130 HZ). THAT IS WHY WHEN MEASURING BETWEEN 10 AND 310 HZ, 3 PEAKS ARE OBSERVED ON THE BODE PLOT (32HZ, 120HZ, 247HZ)
- AS WITH THE TESTING FOR THE CANTILEVER BEAM, THE SINE SWEEP PRODUCES CLEARER BODE PLOT PEAKS BUT TAKES MUCH LONGER (3 MINUTES, 47 SECONDS). THE RANDOM NOISE TEST HAS LESS SHARPER PEAKS BUT IS OVERALL FASTER (1 MINUTE, 30 SECONDS).
- THE WING HAS HIGHER BENDING NATURAL FREQUENCIES THAN THE BEAM.
  - THE CROSS SECTION IS LARGER NEAR THE INSIDE, INCREASING RIGIDITY AND STIFFNESS.
  - THE MASS IS ALSO MORE CONCENTRATED INWARDS. THEREFORE, THE END IS LIGHTER WHICH CONTRIBUTES TO A HIGHER WN. $\omega_n = \sqrt{\frac{k}{m}}$

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CHECKLIST

Task Description	X
HARDWARE SETUP COMPLETED AS PER FIGURE 1	X
COMPLETE TABULAR COLUMN FOR SINE SWEEP TEST 1	X
COMPLETE TABULAR COLUMN FOR SINE SWEEP TEST 2	X
OBTAIN VIDEOS V1, V2, & V3	X
COMPLETE TABULAR COLUMN FOR RANDOM NOISE TEST 1	X
COMPLETE TABULAR COLUMN FOR RANDOM NOISE TEST 2	X
OBTAIN VIDEOS V4, V5, & V6	X
POWER OFF SPECTRUM ANALYZER	X
TURN OFF POWER TO THE SHAKER AMPLIFIER (PUSH BUTTON ON FRONT PANEL)	X
UNPLUG THE TWO POWER CORDS FOR THE SHAKER AMPLIFIER	X
TURN OFF POWER TO THE SIGNAL CONDITIONER (TOGGLE SWITCH)	X

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